AMS526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 22: Conjugate Gradient Method

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Outline

Krylov Subspace Algorithms

- Create a sequence of Krylov subspaces for $Ax = b$

$$\mathcal{K}_k = \{b, Ab, \ldots, A^{k-1}b\}$$

  and find an "optimal" solutions x_k in $\mathcal{K}_k$ at kth step
- Only matrix-vector products involved
- For SPD matrices, the most famous method is *Conjugate Gradient* (CG) method discovered by Hestenes/Stiefel in 1952
  - Finds best solution $x_k \in \mathcal{K}_k$ in norm $\|x\|_A \equiv \sqrt{x^{\mathrm{T}}Ax}$
  - Only requires storing 4 vectors (instead of k vectors) due to three-term recurrence

Motivation of Conjugate Gradients

- If $A \in \mathbb{R}^{n \times n}$ is SPD, then quadratic function

$$\varphi(x) = \frac{1}{2} x^{\mathrm{T}} A x - x^{\mathrm{T}} b$$

  has unique minimum
- Negative gradient of this function is residual vector

$$-\nabla \varphi(x) = b - Ax = r$$

  so minimum is obtained precisely when $Ax = b$
- Optimization methods have form

$$x_{k+1} = x_k + \alpha_k p_k$$

  where p_k is *search direction* and α is *step length* chosen to minimize $\varphi(x_k + \alpha_k p_k)$
- Line search parameter is $\alpha_k = r_k^{\mathrm{T}} p_k / p_k^{\mathrm{T}} A p_k$
- In CG, p_k is chosen to be A-conjugate (or A-orthogonal) to previous search directions, i.e., $p_k^{\mathrm{T}} A p_j = 0$ for $j < k$

Conjugate Gradient Method

Algorithm: Conjugate Gradient Method

$x_0 = 0,\ r_0 = b,\ p_0 = r_0$

for $k = 1, 2, 3, \ldots$

$\alpha_k = (r_{k-1}^{\mathrm{T}} r_{k-1})/(p_{k-1}^{\mathrm{T}} A p_{k-1})$	step length
$x_k = x_{k-1} + \alpha_k p_{k-1}$	approximate solution
$r_k = r_{k-1} - \alpha_k A p_{k-1}$	residual
$\beta_k = (r_k^{\mathrm{T}} r_k)/(r_{k-1}^{\mathrm{T}} r_{k-1})$	improvement this step
$p_k = r_k + \beta_k p_{k-1}$	search direction

- Only one matrix-vector product Ap_{k-1} per iteration
- Apart from matrix-vector product, #flops per iteration is $O(n)$
- If A is sparse with constant number of nonzeros per row, $O(n)$ operations per iteration
- CG can be viewed as minimization of quadratic function $\varphi(x) = \frac{1}{2}x^{\mathrm{T}}Ax - x^{\mathrm{T}}b$ by modifying steepest descent

An Alternative Interpretation of CG

Algorithm: CG

$x_0 = 0,\ r_0 = b,\ p_0 = r_0$
for $k = 1, 2, 3, \ldots$
$\alpha_k = (r_{k-1}^{\mathrm{T}} r_{k-1})/(p_{k-1}^{\mathrm{T}} A p_{k-1})$
$x_k = x_{k-1} + \alpha_k p_{k-1}$
$r_k = r_{k-1} - \alpha_k A p_{k-1}$
$\beta_k = (r_k^{\mathrm{T}} r_k)/(r_{k-1}^{\mathrm{T}} r_{k-1})$
$p_k = r_k + \beta_k p_{k-1}$

Algorithm: A non-standard CG

$x_0 = 0,\ r_0 = b,\ p_0 = r_0$
for $k = 1, 2, 3, \ldots$
$\alpha_k = r_{k-1}^{\mathrm{T}} p_{k-1}/(p_{k-1}^{\mathrm{T}} A p_{k-1})$
$x_k = x_{k-1} + \alpha_k p_{k-1}$
$r_k = b - A x_k$
$\beta_k = -r_k^{\mathrm{T}} A p_{k-1}/(p_{k-1}^{\mathrm{T}} A p_{k-1})$
$p_k = r_k + \beta_k p_{k-1}$

- The non-standard one is less efficient but easier to understand
- It is easy to see $r_k = r_{k-1} - \alpha_k A p_{k-1} = b - A x_k$
- We need to show:
  - α_k minimizes φ along search direction p_k
  - α_k and β_k are equivalent to those in standard CG
  - Minimizing φ along p_k also minimizes φ within Krylov subspace

Optimality of Step Length

- Select step length α_k over vector p_{k-1} to minimize $\varphi(x) = \frac{1}{2}x^{\mathrm{T}}Ax - x^{\mathrm{T}}b$
- Let $x_k = x_{k-1} + \alpha_k p_{k-1}$,

$$\begin{aligned}\varphi(\alpha_k) &= \frac{1}{2}(x_{k-1} + \alpha_k p_{k-1})^{\mathrm{T}} A(x_{k-1} + \alpha_k p_{k-1}) - (x_{k-1} + \alpha_k p_{k-1})^{\mathrm{T}} b \\ &= \frac{1}{2}\alpha_k^2 p_{k-1}^{\mathrm{T}} A p_{k-1} + \alpha_k p_{k-1}^{\mathrm{T}} A x_{k-1} - \alpha_k p_{k-1}^{\mathrm{T}} b + \text{constant} \\ &= \frac{1}{2}\alpha_k^2 p_{k-1}^{\mathrm{T}} A p_{k-1} - \alpha_k p_{k-1}^{\mathrm{T}} r_{k-1} + \text{constant}\end{aligned}$$

- Therefore,

$$\frac{d\varphi}{d\alpha_k} = 0 \Rightarrow \alpha_k p_{k-1}^{\mathrm{T}} A p_{k-1} - p_{k-1}^{\mathrm{T}} r_{k-1} = 0 \Rightarrow \alpha_k = \frac{p_{k-1}^{\mathrm{T}} r_{k-1}}{p_{k-1}^{\mathrm{T}} A p_{k-1}}.$$

- In addition, $p_{k-1}^{\mathrm{T}} r_{k-1} = r_{k-1}^{\mathrm{T}} r_{k-1}$ because $p_{k-1} = r_{k-1} + \beta_k p_{k-2}$ and $r_{k-1}^{\mathrm{T}} p_{k-2} = 0$ due to the following theorem.

Outline

Krylov Subspace in Conjugate Gradient

Theorem (Theorem 38.1 in NLA p. 295)

If $r_{k-1} \neq 0$, spaces spanned by approximate solutions x_k, search directions p_k, and residuals r_k are all equal to Krylov subspaces

$$\begin{aligned}\mathcal{K}_k &= \langle x_1, x_2, \ldots, x_k \rangle = \langle p_0, p_1, \ldots, p_{k-1} \rangle \\ &= \langle r_0, r_1, \ldots, r_{k-1} \rangle = \langle b, Ab, \ldots, A^{k-1} b \rangle\end{aligned}$$

The residuals are orthogonal (i.e., $r_k^T r_j = 0$ for $j < k$) and search directions are A-conjugate (i.e, $p_k^T A p_j = 0$ for $j < k$).

This theorem implies that

$$\alpha_k = (r_{k-1}^{\mathrm{T}} r_{k-1})/(p_{k-1}^{\mathrm{T}} A p_{k-1}) = r_{k-1}^{\mathrm{T}} p_{k-1}/(p_{k-1}^{\mathrm{T}} A p_{k-1})$$

and

$$\beta_k = \frac{r_k^{\mathrm{T}} r_k}{r_{k-1}^{\mathrm{T}} r_{k-1}} = \frac{r_k^{\mathrm{T}}(r_{k-1} - \alpha_k A p_{k-1})}{r_{k-1}^{\mathrm{T}} r_{k-1}} = -\frac{r_k^{\mathrm{T}} A p_{k-1}}{p_{k-1}^{\mathrm{T}} A p_{k-1}}.$$

Proof of Properties of CG

Prove based on notation of standard CG.

- Proof of equality of subspaces by simple induction.
- To prove $r_k^{\mathrm{T}} r_j = 0$, note that $r_k = r_{k-1} - \alpha_k A p_{k-1}$ and $(Ap_{k-1})^{\mathrm{T}} = p_{k-1}^{\mathrm{T}} A$, so

$$r_k^{\mathrm{T}} r_j = \left(r_{k-1} - \alpha_k A p_{k-1}\right)^{\mathrm{T}} r_j = r_{k-1}^{\mathrm{T}} r_j - \alpha_k p_{k-1}^{\mathrm{T}} A r_j.$$

  - If $j < k-1$, then both terms on right are zero by induction.
  - If $j = k-1$, plug in $\alpha_k = (r_{k-1}^{\mathrm{T}} r_{k-1})/(p_{k-1}^{\mathrm{T}} A p_{k-1})$

$$r_{k-1}^{\mathrm{T}} r_j - \alpha_k p_{k-1}^{\mathrm{T}} A r_j = r_{k-1}^{\mathrm{T}} r_{k-1} - r_{k-1}^{\mathrm{T}} r_{k-1} \frac{p_{k-1}^{\mathrm{T}} A r_{k-1}}{p_{k-1}^{\mathrm{T}} A p_{k-1}},$$

  which is zero because

$$p_{k-1}^{\mathrm{T}} A p_{k-1} = p_{k-1}^{\mathrm{T}} A (r_{k-1} + \beta_k p_{k-2}) = p_{k-1}^{\mathrm{T}} A r_{k-1}$$

  by induction hypothesis.

Proof Cont'd

- To prove $p_k^{\mathrm{T}} A p_j = 0$, note that $p_k = r_k + \beta_k p_{k-1}$, so

$$p_k^{\mathrm{T}} A p_j = r_k^{\mathrm{T}} A p_j + \beta_k p_{k-1}^{\mathrm{T}} A p_j.$$

  - If $j < k-1$, then both terms on right are zero by induction.
  - If $j = k-1$, plug in $\beta_k = (r_k^{\mathrm{T}} r_k)/(r_{k-1}^{\mathrm{T}} r_{k-1})$,

$$\begin{aligned} r_k^{\mathrm{T}} A p_j + \beta_k p_{k-1}^{\mathrm{T}} A p_j &= r_k^{\mathrm{T}} A p_{k-1} + \frac{1}{\alpha_k} r_k^{\mathrm{T}} r_k \\ &= \frac{1}{\alpha_k} r_k^{\mathrm{T}} (r_k + \alpha_k A p_{k-1}) \\ &= \frac{1}{\alpha_k} r_k^{\mathrm{T}} r_{k-1} \\ &= 0. \end{aligned}$$

Relationship with Lanczos Iteration

CG and Lanczos iteration are essentially the same process

- In CG, let b be right-hand side of $Ax = b$

$$\begin{aligned}\mathcal{K}_k &= \langle x_1, x_2, \ldots, x_k \rangle = \langle p_0, p_1, \ldots, p_{k-1} \rangle \\ &= \langle r_0, r_1, \ldots, r_{k-1} \rangle = \langle b, Ab, \ldots, A^{k-1} b \rangle\end{aligned}$$

- In Lanczos iteration for $A \in \mathbb{R}^{n \times n}$, starting from $q_1 = b/\|b\|$

$$AQ_k = Q_{k+1} \tilde{T}_k, \tag{1}$$

  where $\tilde{T}_k$ is $(k+1) \times k$; Q_k is composed of orthonormal basis of $\mathcal{K}_k$
- If q_1 is a multiple of $r_0 = b$, then q_i will be proportional to r_{i-1}
- In (**??**), $\tilde{T}_k = \begin{bmatrix} \alpha_1 & \beta_1 & & \\ \beta_1 & \alpha_2 & \ddots & \\ & \ddots & \ddots & \beta_{k-1} \\ & & \beta_{k-1} & \alpha_k \\ & & & \beta_k \end{bmatrix}$

Alternative Derivation Based on Lanczos Iteration

- Let $x_k = Q_k y_k$. Then,

$$r_k = b - Ax_k = b - AQ_k y_k = b - Q_{k+1}\tilde{T}_k y_k$$

- Let $Q_k^{\mathrm{T}} r_k = Q_k^{\mathrm{T}}\left(b - Q_{k+1}\tilde{T}_k y_k\right) = 0$ (i.e., $r_k \perp \mathcal{K}_k$), we obtain

$$Q_k^{\mathrm{T}} Q_{k+1}\tilde{T}_k y_k = Q_k^{\mathrm{T}} b$$

  where $Q_k^{\mathrm{T}} Q_{k+1}\tilde{T}_k = T_k$ and $Q_k^{\mathrm{T}} b = \beta e_1$ with $\beta = \|b\|$

- Hence,

$$T_k y_k = \beta e_1 \tag{2}$$

  where $T_k = Q_k^{\mathrm{T}} A Q_k$ is tridiagonal, and is SPD if A is SPD

- It takes $\mathcal{O}(1)$ flops to update Cholesky factorization of T_k and then $\mathcal{O}(k)$ flops to solve (**??**). Resulting algorithm is equivalent to CG

Outline

Termination in Exact Arithmetic

Theorem (Theorem 11.3.1 in MC p. 629)

If k_* is dimension of smallest invariant space that contains r_0, then CG terminates in k_* steps in exact arithmetic.

- A subspace $\mathcal{S}$ is *invariant* w.r.t. to A if for any $v \in \mathcal{S}$, $Av \in \mathcal{S}$

Proof.

$r_0 = b$ can be written as a linear combination of k_* eigenvectors of A, $\{v_1, v_2, \ldots, v_{k_*}\}$, so is $x_* = A^{-1}b$ (since A is diagonalizable).
At step $k = k_*$, $\dim(\mathcal{K}_{k_*}) = k_*$, and $\{v_1, v_2, \ldots, v_{k_*}\}$ form a basis of $\mathcal{K}_{k_*}$, and hence $x_* \in \mathcal{K}_{k_*}$.
If $x_* \in \mathcal{K}_k$ for $k < k_*$, $\dim(\mathcal{K}_k) = k < k_*$, then r_0 would have been contained in a lower-dimensional invariant space. Contradiction. □

- If A has s distinct eigenvalues, CG converges in $\leq s$ iterations.
- With rounding errors, we may not get exact x_* after k_* iterations
- In addition, we may want to terminate sooner than k_* iterations

Optimality of Conjugate Gradients

Theorem (Theorem 38.2 in NLA p. 296)

If $r_{k-1} \neq 0$, then error $e_k = x_* - x_k$ is minimized in A-norm in $\mathcal{K}_k$.

Proof.

Consider arbitrary point $x = x_k - \Delta x \in \mathcal{K}_k$ with error $e = x_* - x = e_k + \Delta x$. So

$$\begin{aligned} \|e\|_A^2 &= (e_k + \Delta x)^{\mathrm{T}} A (e_k + \Delta x) \\ &= e_k^{\mathrm{T}} A e_k + \Delta x^{\mathrm{T}} A \Delta x + 2 e_k^{\mathrm{T}} A \Delta x, \end{aligned}$$

where $e_k^{\mathrm{T}} A \Delta x = r_k^{\mathrm{T}} \Delta x = 0$ because $r_k \perp \mathcal{K}_k$. Since A is SPD, $\|e\|_A^2 \geq \|e_k\|_A^2$ and equality holds iff $\Delta x = 0$. □

- Because $\mathcal{K}_k$ grows monotonically, $\|e_k\|_A$ decreases monotonically
- Note: A-norm is defined as $\|x\|_A = \sqrt{x^{\mathrm{T}} A x}$, assuming A is SPD. It is different from weighted norm $\|x\|_W = \|Wx\|$

Convergence Rate with Rounding Errors

- If A has 2-norm condition number κ, error is bounded by

$$\frac{\|e_k\|_A}{\|e_0\|_A} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k$$

- Proof is based on analysis of matrix polynomials
  - CG minimizes $\|p_k(A)e_0\|_A$ at kth step, with $e_0 = x_*$, where p_k is degree-k polynomial $p_k(x) = 1 + c_1 x + c_2 x^2 + \cdots + c_k x^k$
  - $\|e_k\|_A / \|e_0\|_A \leq \inf_{p_k} \max_\lambda |p_k(\lambda)|$, where λ are eigenvalues of A, which is further bounded using theory of orthogonal polynomials
- $2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \approx 2\left(1 - \frac{2}{\sqrt{\kappa}}\right)^k$ for large κ, so CG takes up to $O(\sqrt{\kappa})$ iterations
- In general, CG performs well with clustered eigenvalues